

Spatial Data II: Spatial Econometrics

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Applied Quantitative Methods II
MA in Social Sciences, Spring 2026

Today's goals

- Understand why spatial autocorrelation violates OLS assumptions
- Diagnose spatial dependence: Moran's I on residuals and LM tests
- Estimate the Spatial Error Model (SEM) and interpret λ
- Estimate the Spatial Lag Model (SLM/SAR) and interpret ρ
- Compute direct and indirect effects after SLM estimation
- Understand estimation methods: ML, IV/2SLS, and why OLS fails
- Choose between SEM, SLM, SDM, and extensions (SAC, SDEM)
- Compare models: LR tests for nested models, AIC for non-nested

Roadmap

The Problem: OLS and Spatial Dependence

Detecting Spatial Autocorrelation

The Spatial Error Model (SEM)

The Spatial Lag Model (SLM/SAR)

Direct and Indirect Effects

The Spatial Durbin Model (SDM)

Estimation Methods

Model Selection and Practical Guidance

Model Relationships and Extensions

Beyond Cross-Sections: Spatial Panel Models

A standard OLS regression

$$y_i = \alpha + \mathbf{x}_i\boldsymbol{\beta} + \varepsilon_i$$

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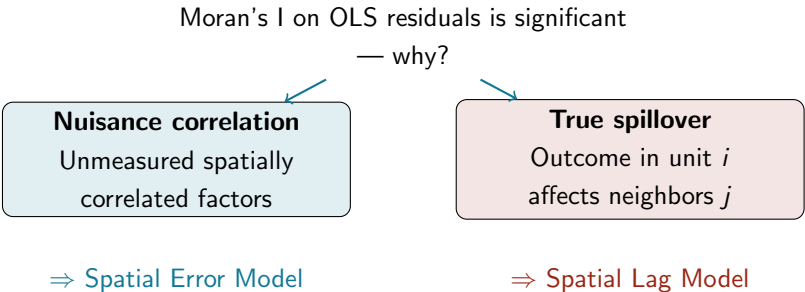
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 - Same problem as with clustered or heteroskedastic errors
 - But the structure of dependence is explicitly *geographic*

Two sources of spatial dependence

Moran's I on OLS residuals is significant

— why?



The diagram illustrates two potential causes for significant Moran's I on OLS residuals. At the top, the text 'Moran's I on OLS residuals is significant' is followed by '— why?'. Two arrows point from this question to two boxes. The left box, titled 'Nuisance correlation', has a light blue background and contains the text 'Unmeasured spatially correlated factors'. Below this box is the text '⇒ Spatial Error Model'. The right box, titled 'True spillover', has a light pink background and contains the text 'Outcome in unit i affects neighbors j '. Below this box is the text '⇒ Spatial Lag Model'.

Nuisance correlation

Unmeasured spatially
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⇒ Spatial Error Model

True spillover

Outcome in unit i
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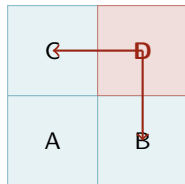
⇒ Spatial Lag Model

Same symptom, different model

In your research: which mechanism is more plausible?

Spatial weights: a brief recap

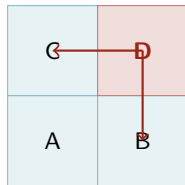
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D's neighbors: B, C

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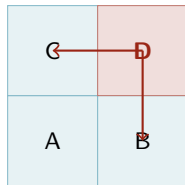
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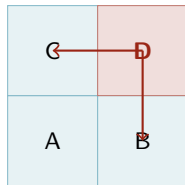
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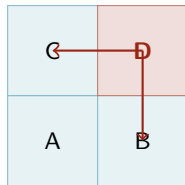
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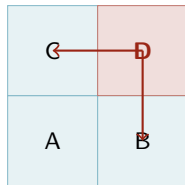
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- In R: `poly2nb()` → `nb2listw()`



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 - High I : similar values cluster together (hot spots and cold spots)
 - Low I : high and low values alternate (checkerboard pattern)

The key diagnostic: Moran's I on residuals

- Run OLS first, compute residuals $\hat{\varepsilon}_i = y_i - \hat{y}_i$

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ols_fit = lm(y ~ x, data = mydata)
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- **LM tests** (Lagrange Multiplier): test for specific types of spatial dependence

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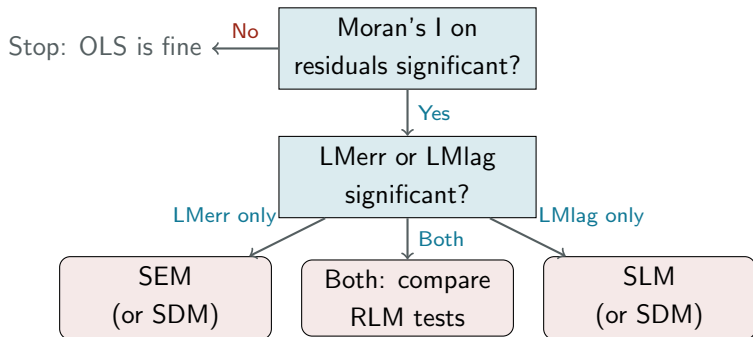
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LM decision rule



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- $\lambda \neq 0$ does **not** mean y affects neighbors' y
 - It means the residuals are spatially structured
- Fix: correct SEs and improve efficiency — not model substantive spillover

SEM in R

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library(spatialreg) # or spdep

# Fit Spatial Error Model
sem_fit = errorsarlm(y ~ x1 + x2,
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summary(sem_fit)
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- `errorsarlm()`: maximum likelihood estimation of SEM

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- `errorsarlm()`: maximum likelihood estimation of SEM
- `zero.policy = TRUE`: allow units with no neighbors
- Key output: $\hat{\lambda}$ and its significance; $\hat{\beta}$ interpreted as OLS

SEM: reading the output

	Estimate	Std. Error	z-value	p-value
(Intercept)	-2.14	0.48	-4.45	< 0.001
gdp_pc	0.63	0.07	9.11	< 0.001
pop_l	0.18	0.05	3.37	0.001
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- Use AIC/LR test to confirm SEM improves on OLS

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- y_i is partly determined by its neighbors' y_j

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- $\rho > 0$: spatial diffusion (neighbors' high y raises your y)
- **Warning**: $\hat{\beta}$ from SLM is **not** the marginal effect of $\mathbf{x}$ on y

SLM in R

```
library(spatialreg)

# Fit Spatial Lag Model
slm_fit = lagsarlm(y ~ x1 + x2,
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summary(slm_fit)
```

- `lagsarlm()`: ML estimation of SLM

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- `lagsarlm()`: ML estimation of SLM
- Key output: $\hat{\rho}$ and its significance; $\hat{\beta}$ (not marginal effects!)
- Do not interpret $\hat{\beta}$ directly — use `impacts()` instead

SLM: reading the output

	Estimate	Std. Error	z-value	p-value
(Intercept)	-1.87	0.52	-3.59	< 0.001
gdp_pc	0.51	0.08	6.29	< 0.001
pop_l	0.14	0.05	2.85	0.004
ρ (Rho)	0.48	0.10	4.90	< 0.001

- $\hat{\rho} = 0.48$: strong positive spatial diffusion in y

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Changing x_i creates a ripple through the network.

The coefficient $\hat{\beta}$ is not the marginal effect.

Why $\hat{\beta}$ is not the marginal effect

Solving the SLM for $\mathbf{y}$:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$(\mathbf{I} - \rho \mathbf{W}) \mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\mathbf{y} = \underbrace{(\mathbf{I} - \rho \mathbf{W})^{-1}}_{\mathbf{B}} (\mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon})$$

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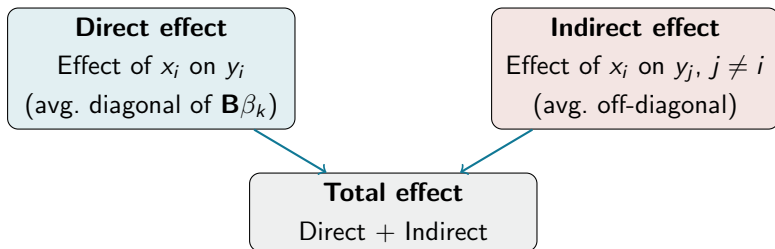
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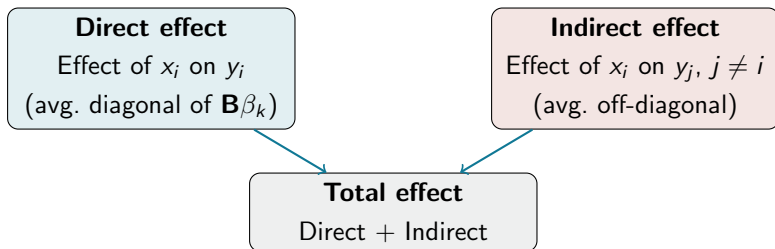
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- **Question:** if you change x_i , how many units in the network are eventually affected?

Direct, indirect, and total effects



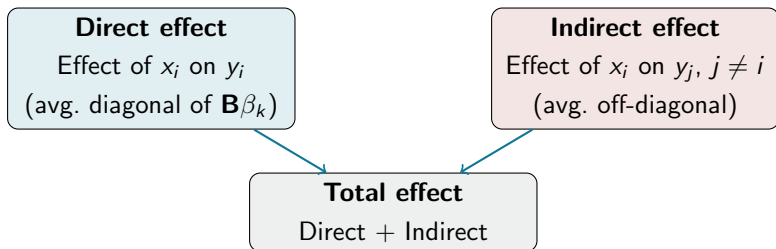
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Direct, indirect, and total effects



- Direct $\approx \hat{\beta}_k$ but slightly larger due to own-unit feedback
- Indirect: captures spatial spillover — ignored by OLS and SEM
- Total: the full equilibrium effect of a uniform shock to x_k

Computing impacts in R

```
# Direct, indirect, total effects
imp = impacts(slm_fit, listw = my_listw)
summary(imp, zstats = TRUE)
```

```
## Impact measures (lag, exact):
```

```
##   Direct Indirect Total
```

```
# gdp_pc 0.574   0.531  1.105
```

```
# pop_l  0.161   0.149  0.310
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- **Direct** (gdp_pc): 0.574 — unit's own GDP raises own y by 0.574

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- Use `zstats = TRUE` to get z-statistics and p -values

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SDM: extending the SLM

Spatial Durbin Model

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$$

- Adds spatially lagged covariates $\mathbf{W}\mathbf{X}\boldsymbol{\theta}$ to the SLM

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 - SLM is nested ($\boldsymbol{\theta} = 0$)
 - SEM is nested ($\boldsymbol{\theta} = -\rho\boldsymbol{\beta}$)
- LeSage & Pace (2009): SDM is often preferred as a conservative default

SDM in R and when to use it

```
# Spatial Durbin Model
sdm_fit = lagsarlm(y ~ x,
  data = mydata,
  listw = my_listw,
  Durbin = TRUE,
  zero.policy = TRUE)
```

```
# Direct, indirect, total as usual
impacts(sdm_fit, listw = my_listw)
```

- Durbin = TRUE: adds **WX** to lagsarlm()

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- Also useful as robustness check for SLM estimates

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How are these models estimated?

Model	Estimation	Why not OLS?
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- Alternative for SEM: **Feasible GLS** or **GMM** (Kelejian & Prucha)

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SEM vs SLM vs SDM: comparison

	SEM	SLM	SDM
What clusters	Errors (u)	Outcome (y)	Both
Spillover in y	No	Yes	Yes
Spillover from x	No	No	Yes
$\hat{\beta}$ = marginal effect	Yes	No	No
Use impacts()	No	Yes	Yes
Nuisance / design	Nuisance	Substantive	Both
R function	errorsarlm()	lagsarlm()	lagsarlm(Durbin=TRUE)

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- **Robustness**: estimate all three, report side by side

The full workflow in R

```
library(sf); library(spdep); library(spatialreg)
```

```
# 1. Build weights
```

```
nb = poly2nb(data, queen = TRUE)
```

```
listw = nb2listw(nb, style = "W", zero.policy = TRUE)
```

```
# 2. OLS + diagnostics
```

```
ols = lm(y ~ x1 + x2, data = data)
```

```
moran.test(residuals(ols), listw = listw)
```

```
lm.LMtests(ols, listw, test = c("LMerr", "LMlag",  
  "RLMerr", "RLMlag"))
```

```
# 3. Spatial models
```

```
sem = errorsarlm(y ~ x1 + x2, data, listw)
```

```
slm = lagsarlm(y ~ x1 + x2, data, listw)
```

```
impacts(slm, listw = listw)
```

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 - Always run Moran's I on OLS residuals before estimating spatial models

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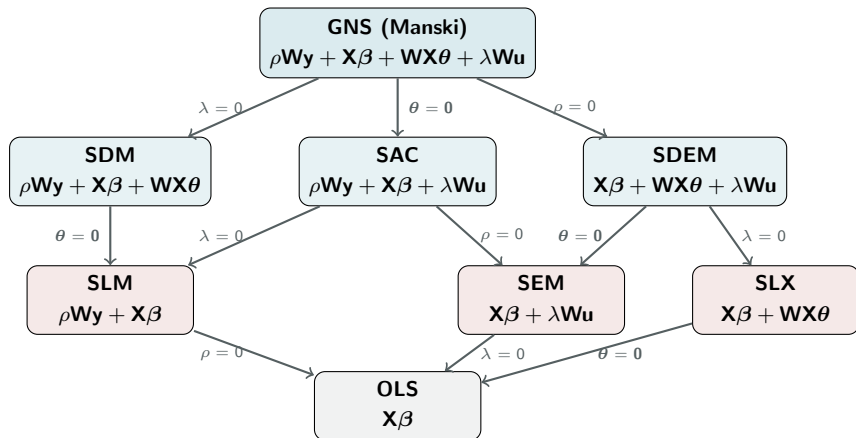
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The family of spatial regression models



Comparing nested models: the LR test

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 - SEM and SLM are nested in SDM, SAC

Comparing nested models: the LR test

- Two models are **nested** if one is a restricted version of the other
 - OLS is nested in SEM ($\lambda = 0$), SLM ($\rho = 0$), SLX ($\theta = \mathbf{0}$)
 - SEM and SLM are nested in SDM, SAC
- **Likelihood Ratio test**: compares log-likelihoods

$$LR = 2[\ell(\text{unrestricted}) - \ell(\text{restricted})] \sim \chi^2(q)$$

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 - OLS is nested in SEM ($\lambda = 0$), SLM ($\rho = 0$), SLX ($\theta = \mathbf{0}$)
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- **Strategy:** use LR tests along nesting paths; use AIC across non-nested alternatives

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Spatial Autoregressive Combined Model

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} = \lambda \mathbf{W}\mathbf{u} + \boldsymbol{\varepsilon}$$

- Combines spatial lag (ρ) and spatial error (λ) in one model

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Summary: nesting and comparison strategy

Comparison	Nested?	Test
OLS vs. SEM	Yes	LR test ($\lambda = 0$)
OLS vs. SLM	Yes	LR test ($\rho = 0$)
SLM vs. SDM	Yes	LR test ($\theta = \mathbf{0}$)
SEM vs. SDM	Yes	LR test ($\theta = -\rho\beta$)
SLM vs. SAC	Yes	LR test ($\lambda = 0$)
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SEM vs. SDEM	Yes	LR test ($\theta = \mathbf{0}$)
SEM vs. SLM	No	AIC / BIC
SDM vs. SAC		AIC / BIC
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Roadmap

The Problem: OLS and Spatial Dependence

Detecting Spatial Autocorrelation

The Spatial Error Model (SEM)

The Spatial Lag Model (SLM/SAR)

Direct and Indirect Effects

The Spatial Durbin Model (SDM)

Estimation Methods

Model Selection and Practical Guidance

Model Relationships and Extensions

Beyond Cross-Sections: Spatial Panel Models

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- Key reference: Elhorst (2014), *Spatial Econometrics: From Cross-Sectional Data to Spatial Panels*
- **Bottom line:** if your data has a spatial *and* temporal dimension, use purpose-built tools — do not improvise with cross-sectional spatial models

For the lab session

- Complete the diagnostics workflow on a provided dataset
 - Run OLS; compute Moran's I on residuals
 - Run LM tests; select the appropriate model
- Estimate SEM and SLM; compare with OLS
 - Interpret $\hat{\lambda}$ and $\hat{\rho}$
 - Use `impacts()` for SLM marginal effects
- Reflect: what mechanism drives spatial dependence in your data?

Questions?